

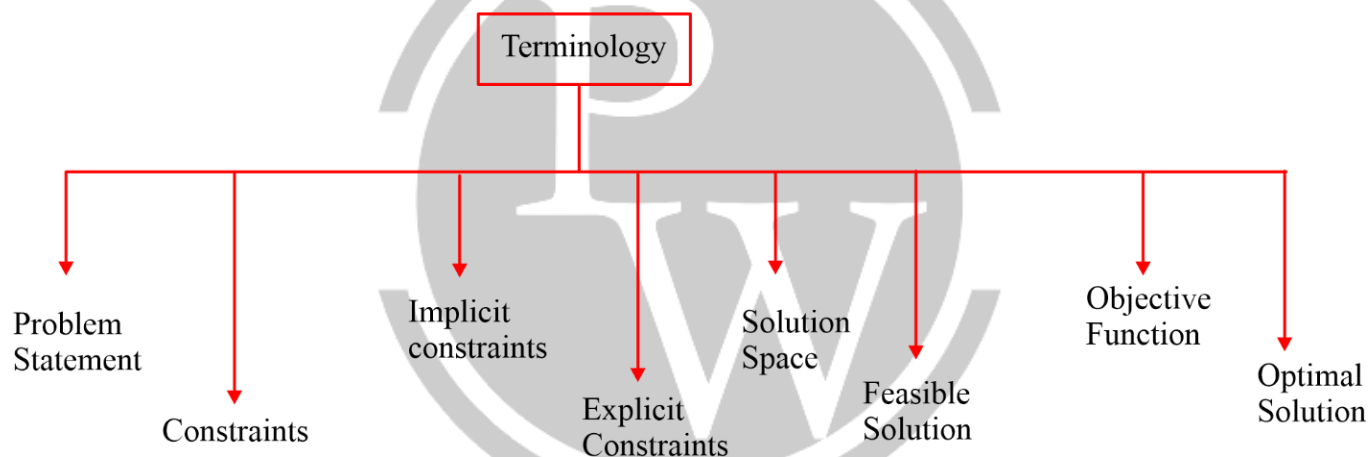
3

GREEDY TECHNIQUE

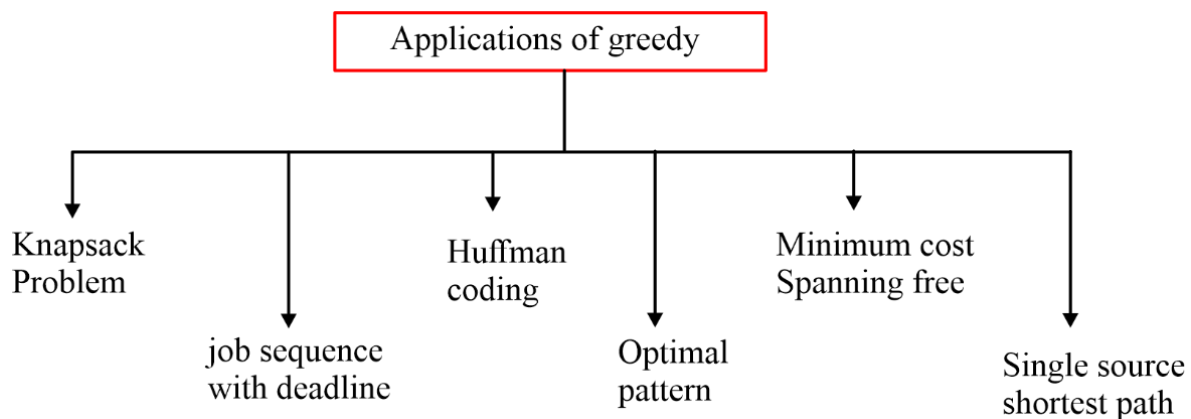
3.1 Greedy Technique

- Greedy method is an algorithm design strategy used for solving problems where solution are seen as result of making a sequence of decisions.
- A problem may contain more than one solution.

3.2 Terminology



3.3 Applications of greedy

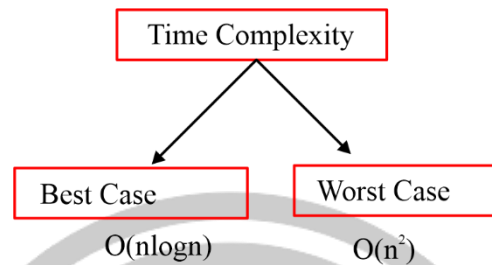


3.4 Knapsack Problem

Time complexity $T(n) = O(n \log n)$

3.5 Job Sequence with Deadline

- Single CPU only.
- Arrival time of each job is same.
- No pre-emption.



3.6 Optimal Merge Pattern

- This is a problem related to merging of files. Given a set of n -files in sorted order. It is required to merge them into a single sorted file with 2-way merging.
- This problem is like merging process in merge sort. In merge sort we were interested in number of comparisons but in optimal merge pattern we are interested in record movement (i.e moving a record from one file to another file).
 - If file F_1 has ' n ' records and file ' F_2 ' has ' m ' records then number of record movement will be ' $m+n$ '. 1 2
- The problem of optimal merge pattern involves merging of n -files ($n \geq 2$).
- At any point choose two records with least weight merge them and put them in list and continue it until all records are merged.
- Time complexity $T(n) = O(n \log n)$
- Space complexity = $O(n)$

3.7 Huffman Coding

- Huffman coding is essentially a non-uniform encoding with convention that the character with higher frequency (probability) of occurrence will be enclosed with less number of bits.
- It comes under data compression technique.
- Time complexity $T(n) = O(n \log n)$